Sinusoidal Steady-State AC Circuits Problems

1.1 Use nodal analysis to find v_o in the circuit of Fig. 1.1.

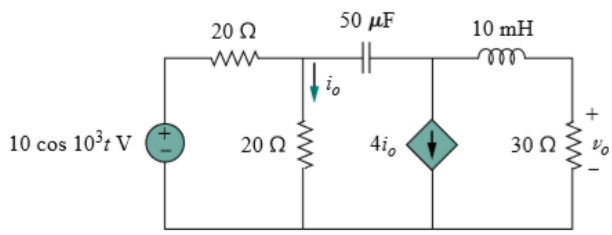

Figure 1.1

1.2 Determine Vo and Io in the circuit of Fig. 1.2 using mesh analysis.

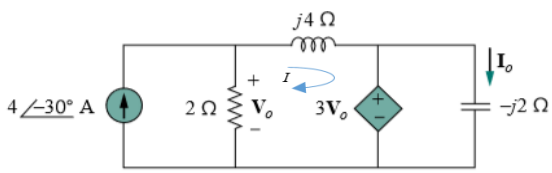

Figure 1.2

1.3 Find io in the circuit in Fig. 1.3 using superposition.

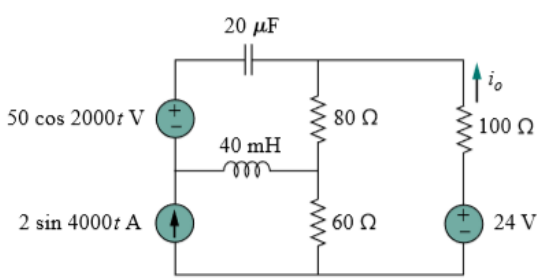

Figure 1.3

1.4 Find the Thevenin equivalent of the circuit in Fig. 1.4 as seen from: (a) terminals a-b (b) terminals c-d.

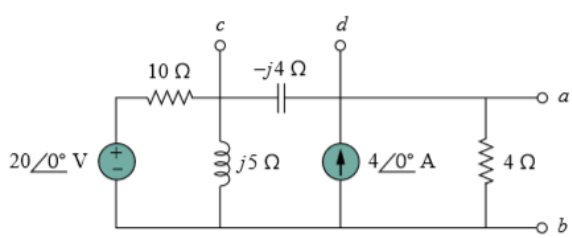

Figure 1.4

1.5 If the input impedance is defined as Zin = Vs/Is, find the input impedance of the op amp circuit in Fig. 1.5 when R1 = 10 kΩ, R2 = 20 kΩ, C1 = 10 nF, C2 = 20 nF, and ω = 5000 rad/s.

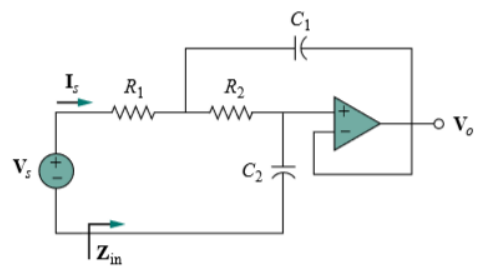

Figure 1.5

1.6 Obtain $v_o(t)$ for the op amp circuit in Fig. 1.6 if $v_s = 4\cos(1000t - 60^\circ)$ V.

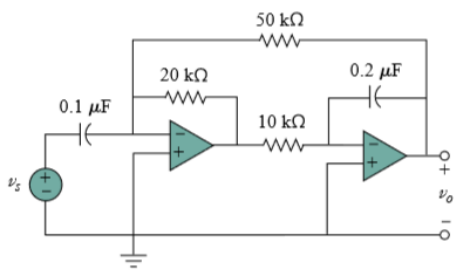

Figure 1.6

1.7 Refer to the circuit depicted in Fig. 1.7. Find the average power absorbed by each element.

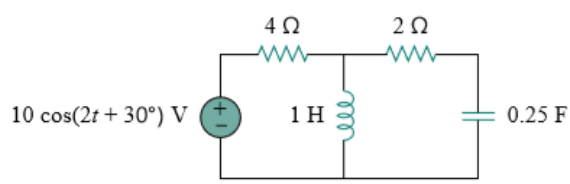

Figure 1.7

1.8 Given the circuit of Fig. 1.8, find the average power absorbed by the 10Ω resistor.

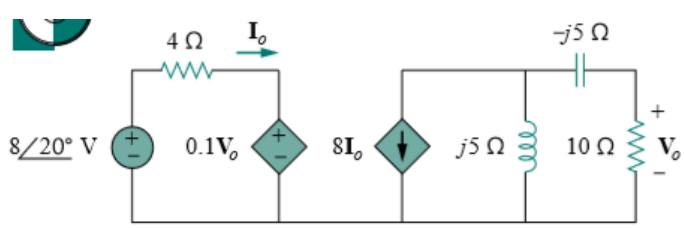

Figure 1.8

1.9 Given the circuit in Fig. 1.9, find Io and the overall complex power supplied.

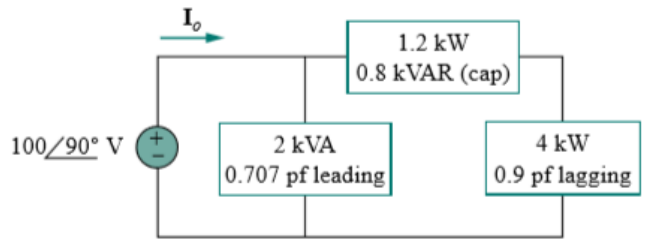

Figure 1.9

1.10 In the circuit of Fig. 1.10, find the value of Z_L that will absorb the maximum power and the value of the maximum power.

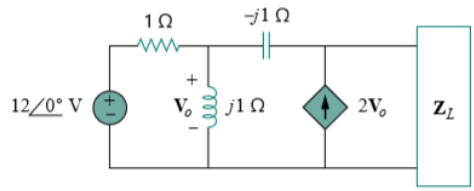

Figure 1.10

1.11 The variable resistor R in the circuit of Fig. 1.11 is adjusted until it absorbs the maximum average power. Find R and the maximum average power absorbed.

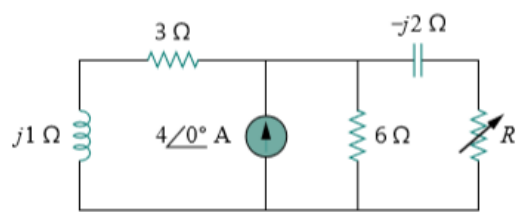

Figure 1.11

1.12 Solve for the line currents in the Y-Δ circuit of Fig.12. Take $Z_\Delta = 60\angle 45\ °\Omega$.

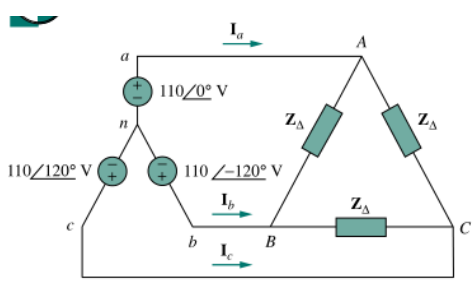

Figure 1.12

1.13 Obtain the line currents in the three-phase circuit of Fig.13.

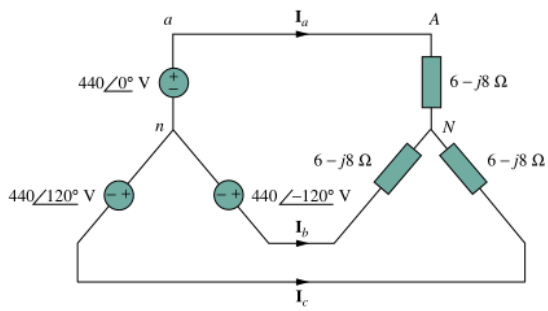

Figure 1.13

1.14. For the three-phase circuit in Fig. 14, find the average power absorbed by the delta-connected load with $Z_\Delta = 21 + j24\Omega$.

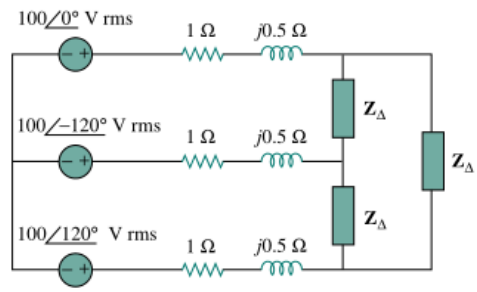

Figure 1.14